

IB Computer Science IA – Guide to Criterion C (System Overview)

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Syllabus Requirement for Criterion C

Criterion C: System overview (6 marks)

The system overview of the product must be consistent with the problem specification in criterion A and the planning in criterion B.

- The system overview should include a system model with the key components, their relationships, the rules governing their interaction, and the algorithms required by these components and the user interface.
- The system overview should have the clarity to enable a third party to recreate the product.
- The system model will provide the information for a viable testing strategy.

The recommended word count for this criterion is **150 words**.

Marks	Level descriptor
0	The response does not reach a standard described by the descriptors below.
1–2	The response: • outlines a limited system model • identifies algorithms for the components of the system model • identifies a testing strategy for at least one success criterion.
3–4	The response: • constructs a system model that is not complete • constructs algorithms for the components of the system model that lead to partial functionality of the product • outlines a testing strategy that aligns with at least three success criteria.
5–6	The response: • constructs a complete system model • constructs algorithms for the components of the system model that enable the product to perform • describes a testing strategy that aligns with the success criteria.

□ Purpose of Criterion C

Criterion C is where you **design and explain your system before (or alongside) coding**. It bridges your **planning (Criterion B)** with your **development (Criterion D)**.

Examiners want to see:

- A **clear model of the system** (so someone else could build it from your design).
- **Algorithms** (pseudocode, flowcharts, or structured natural language).
- A **testing strategy** aligned with success criteria.

This is worth **6 marks** (one of the higher-weighted sections compared to A/B/E).

□ What Criterion C Must Include

Based on the 2027 syllabus:

1. **System Model (Diagrams)**
 - o Show the **components of the system and their relationships**.
 - o Recommended diagrams:
 - Architecture Diagram
 - Data flow diagram (DFD)
 - UML (class diagram, if OOP used)
 - UI navigation Map
 - Entity-Relationship Diagram (ERD, if using databases)
 - User interface mockups / wireframes
2. **Algorithms**
 - o Present algorithms for key features in **pseudocode** or **flowcharts** where implemented later in Criterion D.
 - o **Each major component** should have at least one algorithm.
 - o Example: Book search, borrow/return process, overdue fine calculation.
3. **Testing Strategy**
 - o Create a **table of test cases**.
 - o Cover **normal, boundary, and erroneous input cases**.
 - o Link test cases directly to **success criteria** from Criterion A.

□ “Examiner-friendly” structure you can copy

Criterion C — System Overview (≈150 words)

- **System Model:** (Figure C1: Architecture) (Figure C2: ERD) (Figure C3: UI)
- **Key Algorithms:** (Algorithm 1... Algorithm 2...)
- **Testing Strategy:** (Table C1: Tests aligned to SC1–SCn)

□ How to Score Full Marks (5–6/6 range)

To achieve **full marks in Criterion C**:

- 1 **Construct a complete system model** (covering all features).
- 2 **Present algorithms** that allow your product to function (not just vague outlines).
- 3 **Describe a testing strategy** that clearly maps to the success criteria.
- 4 Diagrams + pseudocode must be **clear enough that a third party could build your system**.

△ Mistakes that lose marks:

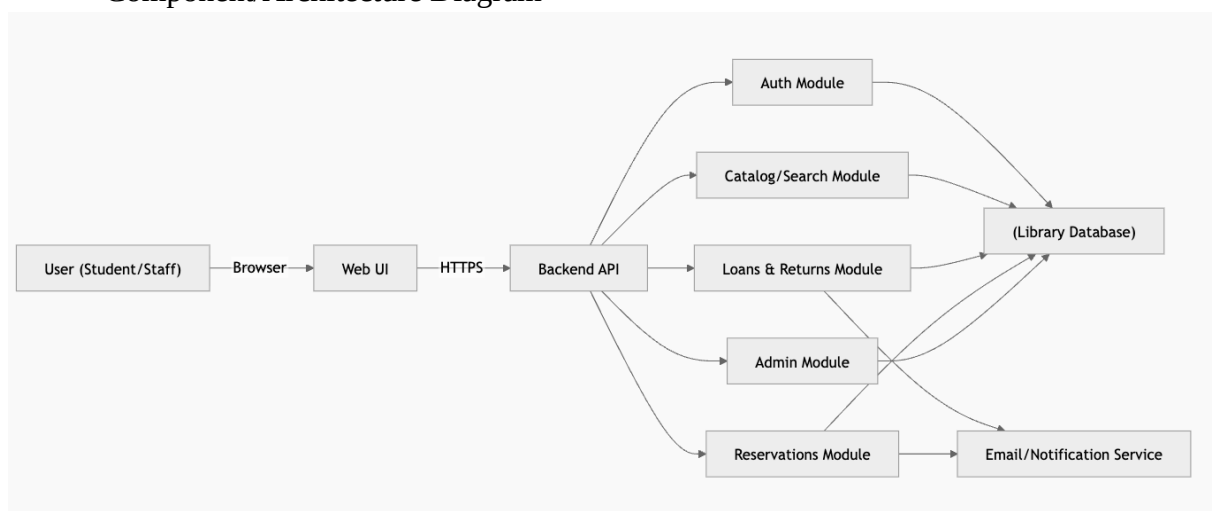
- Vague diagrams (just “input → output”).
- Only describing functionality in words (no pseudocode or flowcharts).
- No link between testing and success criteria.

□ Example: Criterion C (System Overview) – *Library Management System*

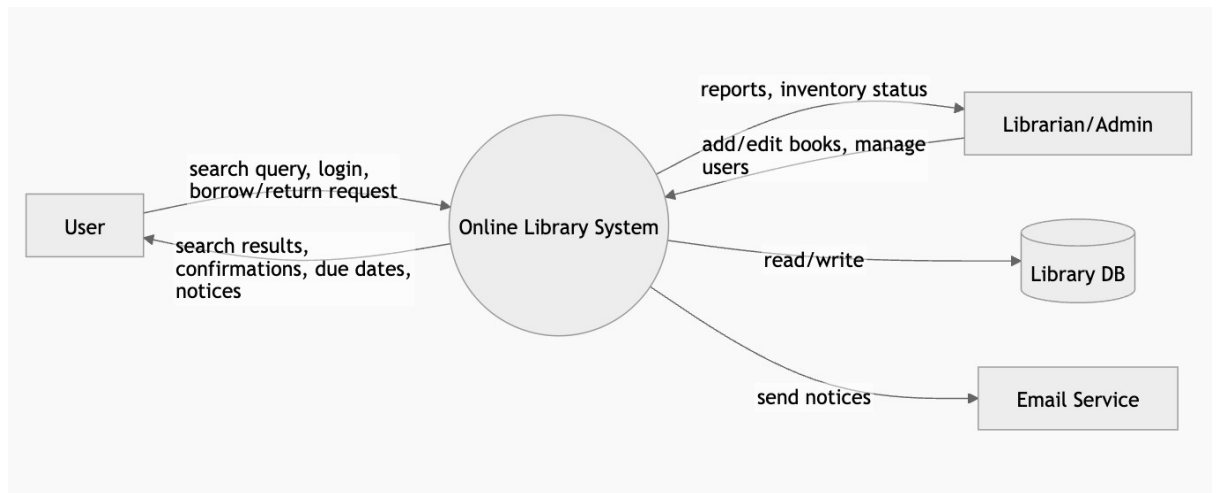
Word Count: ~150 (within 150–200 guideline, excluding diagrams & tables)

0. System model

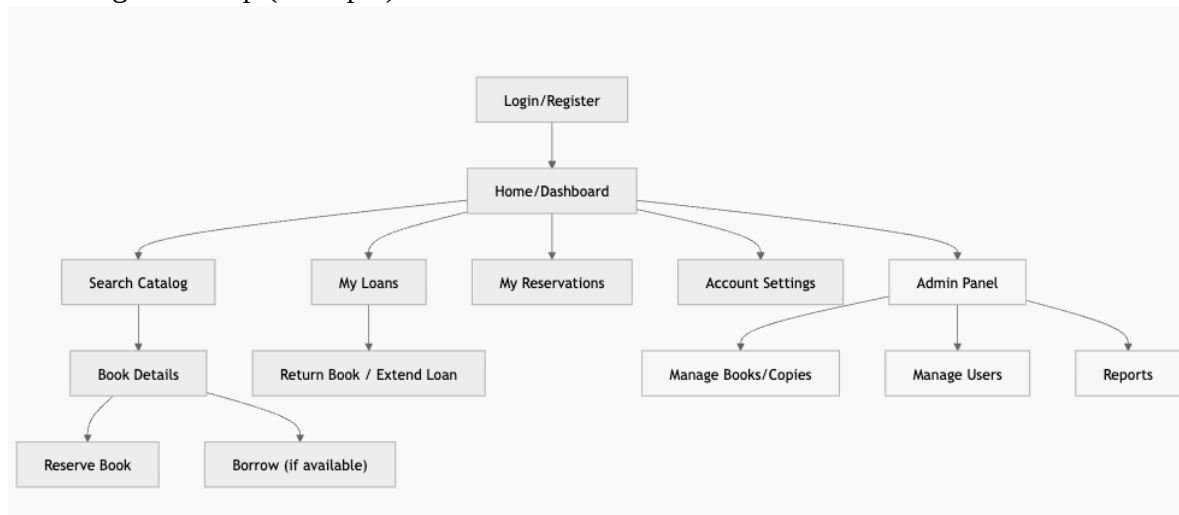
- Component/Architecture Diagram



- DFD Level 0 (Context Diagram)



- UI Navigation Map (example)



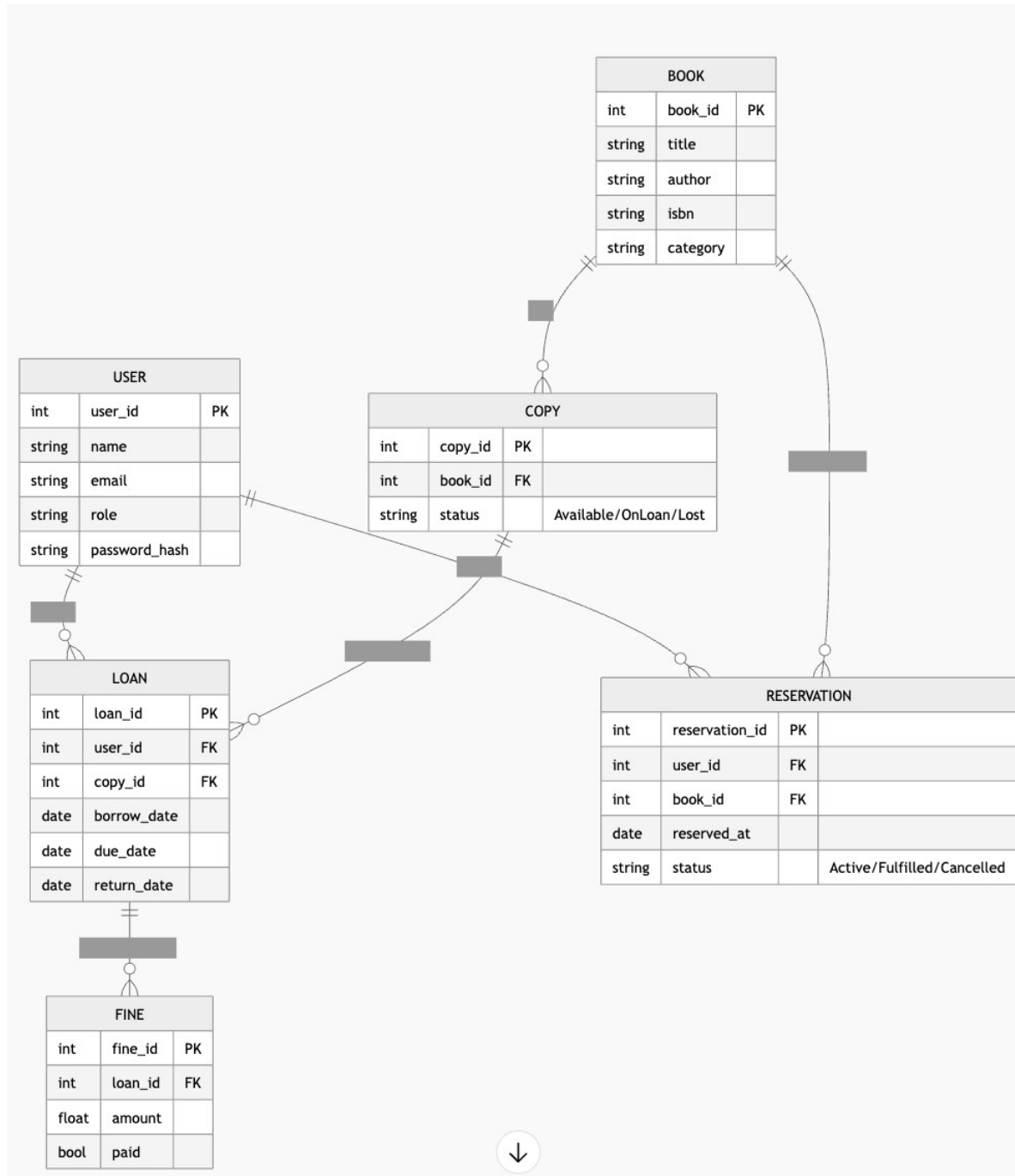
1. Database Scheme

- Tables (example)

Table: Users (stores account details and roles)

Field name	Data type	Key/Constraints	Description / Example
user_id	INT	PK, auto-increment	Unique user ID (e.g., 15)
name	VARCHAR(60)	NOT NULL	Full name ("Alice Chen")
email	VARCHAR(120)	UNIQUE, NOT NULL	Login email
password_hash	VARCHAR(255)	NOT NULL	Hashed password
role	VARCHAR(10)	NOT NULL, DEFAULT 'student'	student/admin

- ERD diagram



2. The Library Management System consists of five main components:

1. **Authentication Module** – verifies librarian/student login using username/password.

- wireframe of GUI
- function description
- algorithm (pseudo/flowchart)

for example

2.3 - WIREFRAME: USER LOGIN

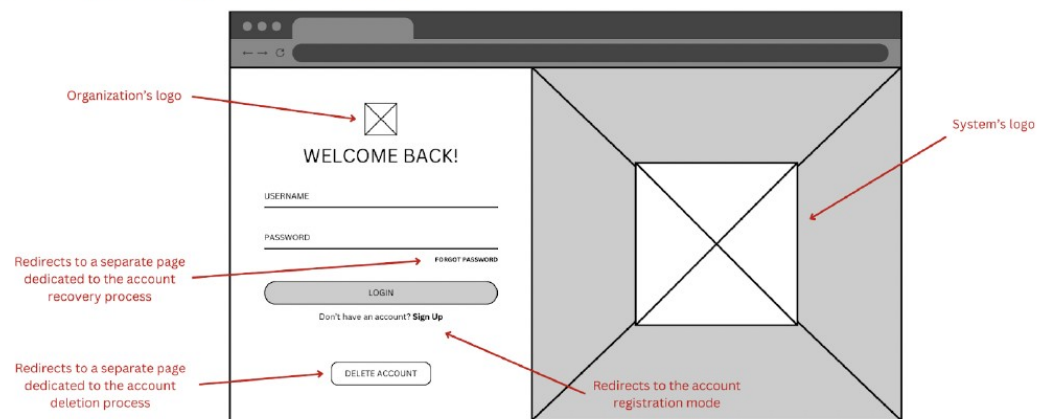


Fig. 2.3.1 - Login Page

The login screen will be the initial point of entry for every user upon their visit to the website. Since both fields are required to successfully identify and verify the user, after the submit button is pressed, the system will first validate that all the fields are filled. In the case of a successful match, the system retrieves the user's device IP address and associates it with the corresponding row in the account database. This allows the system to distinguish between user requests when multiple users are connected simultaneously.

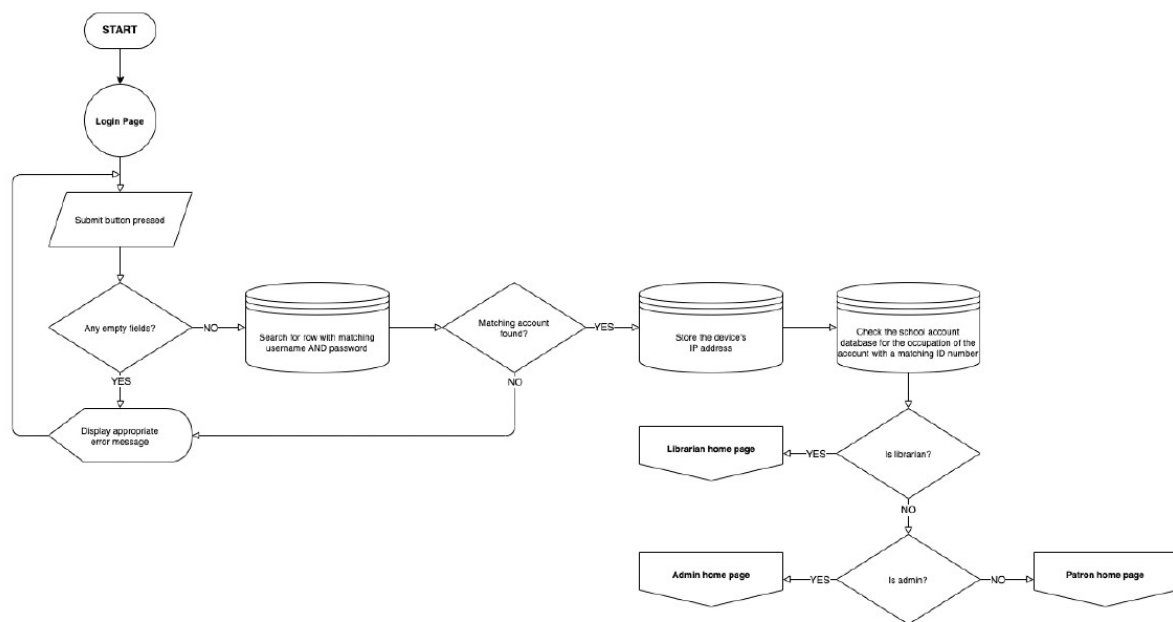


Fig. 2.3.2 - Login Verification Flowchart

2. **Borrow/Return Module** – updates book status, records due dates, calculates fines.

- wireframe of GUI
- function description
- algorithm (pseudo/flowchart)

3. **Search Module** – retrieves books using SQL queries (title, author, ISBN).

- wireframe of the GUI
- function description
- algorithm (pseudo/flowchart)

4. **Reporting Module** – generates overdue book lists and usage statistics.

- wireframe of the GUI
- function description
- algorithm (pseudo/flowchart)

Algorithms Example (pseudo or flowchart)

later match in the Criterion D

Borrow Book Algorithm (pseudocode):

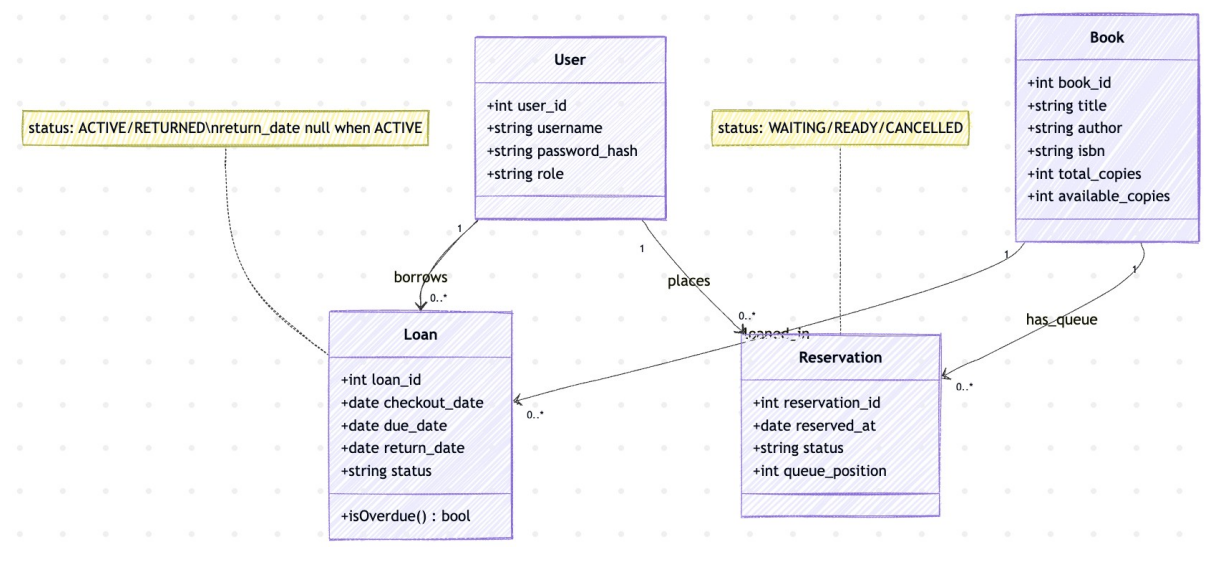
```

FUNCTION borrowBook(userID, bookID):
  IF userID not in registeredUsers:
    RETURN "User not found"
  IF bookID not in availableBooks:
    RETURN "Book not available"
  UPDATE database: set bookID status = "issued"
  SET dueDate = currentDate + 14 days
  RECORD transaction(userID, bookID, dueDate)
  RETURN "Book successfully borrowed"

```

Search Algorithm (flowchart) – included in Figure 3: user input → query database → return matching records → display in GUI.

3. UML Diagram



4. Testing Strategy

Columns like:

Test ID	Linked SC	Test data + steps	Expected result
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Include at least one **invalid case** (e.g., checkout when no copies available, invalid ISBN search, unauthorized admin action).

Test ID	Success Criterion (SC#)	Test (data + steps)	Expected result
T01	SC1 Login works	Data: alice@school.edu, pw Correct!23 Steps: Login → enter details → Login	Dashboard loads; user name shown
T02	SC1 Reject wrong password	Data: alice@school.edu, pw wrong123 Steps: Login → enter → Login	Error message; no login session
T03	SC2 Search by keyword	Data: keyword potter Steps: Search bar → type → search	Relevant books appear in results
T04	SC2 No-results handling	Data: keyword zzzznotabook Steps: Search → submit	“No results found” shown
T05	SC3 Borrow available copy	Data: Copy C102 status=Available Steps: Book details → Borrow	Loan created; copy status becomes OnLoan; due date set
T06	SC3 Block borrow when none available	Data: Book B55 has 0 available copies Steps: Book details → Borrow	Borrow prevented; no loan created
T07	SC3 Return updates records	Data: Loan L201 return_date=NULL Steps: My Loans → Return	return_date saved; copy becomes Available
T08	SC4 Fine calculation	Data: due=2026-02-10, returned=2026-02-13, rate=\$0.50/day Steps: Return overdue book	Fine = $3 \times 0.50 = \$1.50$
T09	SC5 Reserve when all on loan	Data: Book B12 all copies OnLoan Steps: Book details → Reserve	Reservation created; appears in My Reservations
T10	SC6 Non-admin blocked from admin	Data: student account Steps: open /admin	Access denied or redirected
T11	SC6 Admin adds a new book	Data: Title “Atomic Habits”, ISBN 9780735211292, copies=3 Steps: Admin → Add book → Save	Book appears in catalog; 3 copies created Available

This testing strategy ensures all success criteria are verifiable.

This would score **6/6** because:

- **System model** is complete (all modules, relationships).
- **Algorithms** are clear (pseudocode + flowchart).
- **Testing strategy** is systematic and tied to success criteria.